

Gaurav Bhasin

<https://github.com/gbhasin0828?tab=repositories>

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Data Science & MLOps Leader to deliver Large Language & other AI capabilities | My work help conceptualize how to utilize AI Python libraries like TensorFlow, PyTorch, HuggingFace & Langchain to develop & deploy AI applications by using customized MLOps capabilities. I enabled Apple, NFL, Mars Wrigley, Kellogg's, & Kimberly-Clark to utilize technologies like Transformers, Cloud Computing to manage full software lifecycle that drove automation & data driven decision making for Performance Marketing.

Data Science (NLP / LLM / Conversational AI)

- Developed custom trained **LLM's** using open-source **transformers** like **BERT, Llama2, & other Hugging Face open transformers** through partial training (**LORA, QLORA**) along with **RAG, Semantic search** to deliver **conversation AI** interface.
- Delivered efficient and low-latency LLM applications through an integrated infrastructure leveraging **embedding generation**, storage, and retrieval using **Pinecone**, along with **transformers-based architectures** for greater contextual understanding.
- Designed and implemented **LLM pipelines** by utilizing **TensorFlow** for efficient training and deployment of models and **PyTorch** for experimentation and fine-tuning of transformer-based architectures.
- Developed Gen AI capabilities by fine-tuning transformers that can understand business context, exercise objectives, & use relevant data articles by integrating NLU with **domain-specific knowledge graphs, RAG, & vector search**. The system leveraged **semantic embeddings stored in a vector database** (e.g., Pinecone, FAISS) to dynamically retrieve data. Additionally, the model was fine-tuned using **reinforcement learning with human feedback (RLHF)** to improve response accuracy and relevance.
- Developed methodologies to perform **Prompt Engineering** for varied data sources like Structure, Text, Power Points, etc. This has been very critical for preparing inputs to partially fine tune LLM using PEFT (LORA, QLORA). Implemented techniques like PyMuPDF, pdfplumber, PDFMiner, PyMuPDF, pdfplumber, PDFMiner, Whisper to extract and prepare data for LLM training.

AIOps / MLOps & Cloud Capabilities

- Partnered with cross functional teams like IT, Data Science, Data Engineering to establish customized **MLOps** process specifically tailored to meet project / application needs.
- Spearheaded the automation of model development, selection & deployment through capabilities like **PyTest** integrated with **MLflow**. The process was automated via CI CD workflows to eliminate manual effort & deployment time.
- Led the automation of data ingestion & AI/ML model development within a **microservice** architecture by automating pipelines using **Airflow & Kubeflow**. This streamlined workflow management, enabled robust testing, and facilitated seamless model enhancements.
- Engineered robust deployment solutions using **Kubernetes** and **EKS autoscaling**, enabling automatic infrastructure scaling based on workload demand, which reduced operational costs and enhanced model performance through efficient resource utilization.
- Transformed monolithic AI/ML models into a **microservices-based architecture**, improving system uptime, latency, & resource efficiency. Leveraged **Docker** to modularize components, aligning workflows with application architecture to drive operational agility.
- Led the custom creation of integration, tracking, & versioning using Data Version Control (**DVC**) framework within the end-to-end ML lifecycle. Strategically implemented **CI/CD** workflows to enhance efficiency, minimize manual intervention, & accelerate updates.
- Led the deployment of ML models with **Flask / FastAPI**, designing a high-performance, scalable serving architecture. This enabled seamless API integration for real-time inference, optimizing latency and dynamically scaling based on demand.
- Spearheaded adoption of **AWS** services like **EC2, ECR, EKS & SageMaker** to architect scalable, cost-efficient ML workflows to dynamically allocate resources, streamline deployment, & enhance model performance.
- Spearheaded the development of ETL pipelines leveraging **Snowflake, Redshift**, etc., to enable seamless data flow for applications. The pipeline's multi-cluster architecture enabled concurrent data loading & querying at scale. Also introduced advanced optimization techniques, like intelligent partitioning, materialized views which significantly enhanced efficiency, scalability, & data availability.

JUN 2023 – PRESENT

AI / ML Team Lead - MARCOMM | Apple | Chicago, IL (Contractor)

Led a team of AI/ML Engineers, and MLOps experts to develop an LLM capability which would automate the reporting, insight generation and next best actions to improve web engagement for customers on apple.com's marketing pages.

MLOps & Engineering:

Championed the entire MLOps lifecycle, from conceptual design to enterprise deployment, driving cross-functional collaboration between Data Science, Engineering to deliver production-ready AI/ML solutions with improved speed, reliability, and reduced operational costs.

- Spearheaded the delivery of scalable ETL pipelines using **Spark, Airflow**, ensuring data processing for enterprise-scale AI/ML apps.
- Developed data integrity and standards for Marketing Communications function by integrating TensorFlow Data Validation (**TFDV**), establishing enterprise-wide data quality benchmarks through automated anomaly detection and schema validation.
- Designed & implemented customized CI/CD workflows, driving automation in data science pipelines. This reduced manual effort and development time, enabling automation, faster debugging, and improved efficiency in model development.

Data Science

- Designed & implemented an **LLM solution** to automate web performance reporting as an alternative to **Adobe Analytics & Tableau**. LLM leveraged **Snowflake tables** to streamline data & eliminate users to have deep knowledge of tables, **eVars**, or **props**. This significantly reduced manual workload, improved accuracy, & generation of reports efficiently with minimal technical expertise.
- Developed **Recommender Systems** to improve online shopping experience. This was developed using a Two Tower methodology to generate personalized recommendations based on consumer & engagement attributes & further fine-tuned using Feature Engineering.
- Implemented **RLHF** to optimize product descriptions, refining content quality. This was achieved through a reward model trained on web performance metrics to guide LLM fine-tuning. Improved content through human feedback on clarity, tone & persuasiveness.

FEB 2021 – JUN 2023

AI & Data Engineering Technical Partner – (Client - Mars Wrigley Company) | Tiger Analytics | Libertyville, IL

Developed Visualization, Data Science & Data Engineering consulting practice & executed projects worth ~\$10m per year by transforming 60+ member teams to deliver high performance automated ML capabilities, thought leadership, & stakeholder management.

Data Science

- Delivered "**Agile Marketing Mix**" models by integrating **econometric modeling** with **multi-touch attribution (MTA)** to provide actionable insights aligned with media planning. Enabled a holistic view across the entire digital funnel, from awareness to conversion, delivering granular insights that enhanced media planning and optimization based on brand and media strategies & goals.
- Automated **Business reporting** by integrating **RAG, semantic search** with **open-source transformers**, enabling **Conversational AI** capability. This solution significantly reduced labor time, enhanced output quality through contextually relevant responses.
- Delivered solutions for **Marketing Mix, Assortment Optimization, Price Elasticity Measurement, E-commerce & Trade Promotions Optimization**. Leveraged techniques such as **Hierarchical Bayesian, XGBoost, & ElasticNet** to provide actionable, data-driven insights. These solutions enhanced decision-making, optimized resource allocation, & drove measurable business impact.

Data Engineering

- Directed data engineers to design scalable **data models** for automating insights from semi & unstructured data sources. Utilized **NLP** techniques and tools like **TensorFlow, spaCy** for preprocessing, feature extraction, & analysis, enabling actionable insights.
- Spearheaded the development of **Node.js**-based dashboards to monitor supplier health, incorporating real-time data ingestion, visualization, and alerting using technologies like **Django, MongoDB, and D3.js**. This innovative capability was highlighted by Mars at the **United Nations**, demonstrating the application of data engineering techniques to promote supply chain transparency & resilience.

SEP 2018 – FEB 2021

Sr Manager - Data Driven marketing | Kimberly Clark | Chicago, IL

- Stood up Retail media analytics capability by driving interoperability across channels like paid search & social programmatic & walled gardens. This optimized consumer journeys by connecting touchpoints across full digital funnel.
- Designed analytics methodologies & capabilities to migrate trade and shopper programs from mass promotions to mobile platforms like Fetch, driving higher-impact programs at lower costs.
- Facilitated collaboration between technology providers like Fetch, Retail Media and CPGs to enrich 1st party data, enabling smarter segmentation and doubling the impact of performance marketing programs.

NOV 2015 – SEP 2018

Manager – Consumer Insights & Business Analytics | S.C Johnson & SON | Racine, WI

- Developed quantitative research techniques like Conjoint, Discrete Choice to enable innovation plan development.
- Developed analytic process to determine Demand Spaces along with its key drivers of choice. The tool is guiding company's growth efforts, from innovation to expansion, from developing consumer communication strategy to executing promotions at retail.

Master of Business Administration | Michigan State University | East Lansing, MI

Computer Engineering | Indian Institute of Technology IIT (BHU) | India